Note 6.4 Hermitian Matrices

Complex Inner Products

For the definition of norm of complex vector, we define

$$\bar{\mathbf{z}}^T = \mathbf{z}^H, ||\mathbf{z}|| = (\mathbf{z}^H\mathbf{z})^{1/2}$$

Definition of Inner Vector over Complex Numbers

Inner product is an operation on a vector space V that corresponds every pair of vectors $\mathbf{w}$ and $\mathbf{z}$ in V, denoted by $\langle \mathbf{z}, \mathbf{w} \rangle$, satisfying that

- The inner product of two identical vector are nonnegative.
- $\langle \mathbf{z}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{z} \rangle}$ (not that it is not commutative law)
- linear combination.

And we define an inner product on $\mathbb{C}^n$ by

$$\langle \mathbf{z}, \mathbf{w} \rangle = \mathbf{w}^H\mathbf{z} = \overline{\mathbf{z}^H\mathbf{w}}$$

Hermitian Matrices

For the matrix with complex elements, the conjugate of the matrix M is $\bar{M}$ that conjugating each of the entries.

And we define $M^H = \overline{M}^T$ and it has the following properties:

- $(A^H)^H = A$
- $(\alpha A + \beta B)^H = \bar{\alpha}A^H + \bar{\beta}B^H$
- $(AC)^H = C^H A^H$

Definition of Hermitian Matrix

A matrix M is Hermitian if $M = M^H$.

And we can know that the elements on its diagonal are real.

It is similar to symmetric matrix in real space.

Theorem 6.4.1

The eigenvalues of a Hermitian Matrix are all real and eigenvectors belonging to distinct eigenvalues are orthogonal.

It can be proved by constructing $\alpha = \mathbf{x}^H A\mathbf{x}$.

Definition of Unitary Matrix

An $n \times n$ matrix U is unitary if its column vectors form an orthonormal set in $\mathbb{C}^n$.

It has the properties:

- $U^H U = UU^H = I$
- $U^{-1} = U^H$
  It is similar to orthogonal matrix in real space.

Corollary 6.4.2

If the eigenvalues of a Hermitian matrix is distinct, then there exists a unitary matrix diagonalize the Hermitian matrix.

The column vectors of the unitary matrix can be obtained by constructing the set $\{\mathbf{u}_1, \cdots, \mathbf{u}_n\}$ where $\mathbf{u}_i = \frac{\mathbf{x}_i}{|\mathbf{x}_i|}$, $\mathbf{x}_i$ is an eigenvector belonging to λ_i for each eigenvalue of A.

Theorem 6.4.3 Schur's Theorem

For an $n \times n$ matrix A, there exists an unitary matrix U such that $U^H AU$ is upper triangular, denoted by T. That is, $U^H AU = T$.

And the factorization $A = UTU^H$ is the Schur decomposition of A.
If A is Hermitian, the matrix T will be diagonal.

Theorem 6.4.4 Spectral Theorem

If A is Hermitian, there exists a unitary matrix U that diagonalizes A.

The Real Schur Decomposition

Definition of Invariant

A subspace S of $\mathbb{R}^n$ is said to be invariant under a matrix A if for each $\mathbf{x} \in S$, $A\mathbf{x} \in S$.

Theorem 6.4.6 The Real Schur Decomposition

If A is an $n \times n$ matrix with real entries, then A can be factored into a product QTQ^T where Q is an orthogonal matrix and T is in Schur form.
Schur form is that

$$T = \begin{pmatrix} B_1 & \times & \cdots & \times \\ & B_2 & & \times \\ & O & \ddots & \\ & & & B_j \end{pmatrix}$$

where the B_i is either 1×1 or 2×2 matrices and each 2×2 block will correspond to a pair of complex conjugate eigenvalues of A.

Corollary 6.4.7 Spectral Theorem--Real Symmetric Matrices

If A is a real symmetric matrix, then there is an orthogonal matrix Q that diagonalizes A; that is, $Q^T AQ = D$, where D is diagonal.

Normal Matrices

Definition

A matrix A is normal if $AA^H = A^H A$.

Theorem 6.4.8

A matrix A is normal if and only if A possesses a complete orthonormal set of eigenvectors.

Complete orthonormal set is the orthonormal set which can construct a set of basis of the column space of A.